PIMPRI CHINCHWAD COLLEGE OF ENGINEERING (PCCOE), PUNE

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

B.Tech Capstone Project Synopsis | Academic Year: 2026 - 2027

Industry Reference: Automotive Engineering AI Reference Solutions (TTL-Classification: Public - Case Study 1)

Project Title:	AUTOSAR HLD Document Analysis Assistant Using Local RAG and Deterministic Verification
Student Name:	Candidate Student Name (PRN: 123B1F001)
Academic Guide:	Prof. Dr. S. K. Kulkarni, Dept of AI & ML, PCCOE
Industry Mentor:	Lead Automotive System Architect, Tata Technologies Ltd.
Domain & Stack:	Automotive AI, AUTOSAR Classic/Adaptive, FastAPI, Streamlit, FAISS, SentenceTransformers

1. Executive Summary & Abstract

Modern Automotive Electronic Control Units (ECUs) are developed based on the AUTOSAR standard. High-Level Design (HLD) documents define Software Components (SWCs), Ports, Interfaces, and Signal matrices across hundreds of pages of unstructured PDFs. Manual verification causes delays, terminology inconsistencies, unmapped ports, and integration failures. This project delivers an enterprise-grade AI assistant that combines context-aware PDF/ARXML ingestion, local FAISS vector search, cited Retrieval-Augmented Generation (RAG), and a deterministic NetworkX graph verification engine to validate interface completeness and revision changes without exposing proprietary code to public cloud services.

2. Objectives

- Ingest and structure complex automotive HLD PDFs and ARXML component specifications.
- Index architectural knowledge into a persistent local FAISS vector store with SentenceTransformers.
- Deliver natural language Q&A; with strict page/section citations and confidence scoring.
- Automatically extract SWC inventories, Port definitions (P-Port, R-Port), and Sender-Receiver interfaces.
- Perform deterministic consistency checks to identify unmapped ports, type mismatches, and revision regressions.
- Provide an intuitive Streamlit UI, FastAPI service layer, human review workflow, and structured audit export.

3. Scope & Technology Position

In Scope	Out of Scope
• Ingestion of HLD PDFs & ARXML catalogs • Local vector embeddings (BGE / MiniLM) • Grounded Q&A; with citations • SWC, Port, Interface extraction • Consistency & revision diff reporting • Human review disposition & PDF export	• Autonomous unreviewed code modifications • Physical ECU silicon flashing • Replacement of formal safety assessor approvals • Cloud transmission of proprietary IP

4. Implementation Milestones & Timeline

Stage	Key Activities	Deliverable
Stage 1 (Weeks 1-4)	Taxonomy, HLD dataset creation, PDF/ARXML parsers, chunking	Ingestion Pipeline

Stage 2 (Weeks 5-8)	Local FAISS vector store, SentenceTransformers, Cited RAG pipeline	Grounded Q&A; Engine
Stage 3 (Weeks 9-12)	NetworkX graph engine, consistency checks, revision diff algorithm	Verification Engine
Stage 4 (Weeks 13-16)	FastAPI backend, Streamlit UI, evaluation benchmarks, packaging	Complete System

Student Signature PRN: 123B1F001	**Project Guide** Prof. Dr. S. K. Kulkarni	**Head of Department** Dept. of AI & ML, PCCOE